

MDM (Master data management)

Q) What is Data management?

Ans) It is a practice of collecting, organizing, protecting & storing an organization's data to perform analyzed for business decisions.

Range:-

$$\max() - \min()$$

Standard deviation:-

1. Find mean

2. Number deviation from the mean

3. Square the numbers

4. Find the sum

5. Find the variance $(n-1)$

6. Square root of the variance

Ex:- 52, 62, 87, 94, 32, 45

1. mean = 62

2. -10, 0, 25, 32, -30, -17

3. 100, 0, 625, 1024, 900, 289

4. 2938

5. $2938 / (n-1) \Rightarrow 2938 / 5 \Rightarrow 587.6$

6. $\sqrt{587.6} \Rightarrow 24.24046$

Associated tables of Base objects :-

1. HMRG (Merge History) Table:- It stores the merge related information. SRC-ROWID-object, TGT-ROWID-object {ROWID-match-Rule.

SRC-ROWID-object (source ROWID object) stores the non-swapping ROWID-object.

TGT-ROWID-object (target ROWID object) stores the swapping ROWID-object.

2. HIST (History) Table:-

* It stores all the history information of the base object record for any changes happening on the base object record.

* The table is created only when the history is enabled for a base object in the MDM Hub Console.

3. HXRF (History of cross reference) Table:-

* It tracks the history of the changes on the xref table for each record.

* The table is created only when the history is enabled for a base object in the MDM Hub Console.

4. HCTL Table:-

* It tracks the history of changes on the trust control table.

* The table is created only when trust is enabled for any columns on the base object, other than enabling the history flag of the base object.

5. VXR table:-

* It is the table to store the details of the validation rules failure at the xref level for each attributes.

6. HVXR Table:-

- * Tracks history of changes on the VXR table.

7. DRTY Table:-

- * Tracks records needed to be tokenized.

Staging table:-

- * Receives data from the landing table via MDM mappings.
- * Between landing & staging, the data is subjected to standardization, cleaning & rejection.
- * Staging table always truncated & loaded.
- * Pkey_SRC_Object is a unique column (primary key) for a staging table.

Associated Tables of Staging tables:-

1. Raw table:-

- * It has same structure of staging table.
- * It maintains the history of the source data & stage data based on the audit trail configuration done in MDM Hub console.

2. PRL table:-

- * It is used to identify of the delta (insert/update).
- * It has same structure of the corresponding landing table.

Reject table:

- * Reject table in Informatica MDM is always maintained along with the staging table.
- * It has structure of the landing table.
- * It stores the data rejects during staging & loading process in MDM.

4. OPL (Online previous Landing) Table:-

- * It is created by MDM SIF services, & stores the cleanse put data if delta detection enabled.

Staging process:-

1. MDM Cleanse engine
2. Data cleansing (By using Functions)
3. Data Standardization (By using 3-Party tools)
4. Data translation
5. Delta Detection
6. Remove Duplicates
7. Hard delete detection (It is a custom factor that can be configured during the stage process to identify & flag records as deleted which are not sent by the source system).

Load process:-

1. Load Insert
2. Load update
3. Load reject
4. Data integrity
5. Data Consolidation.

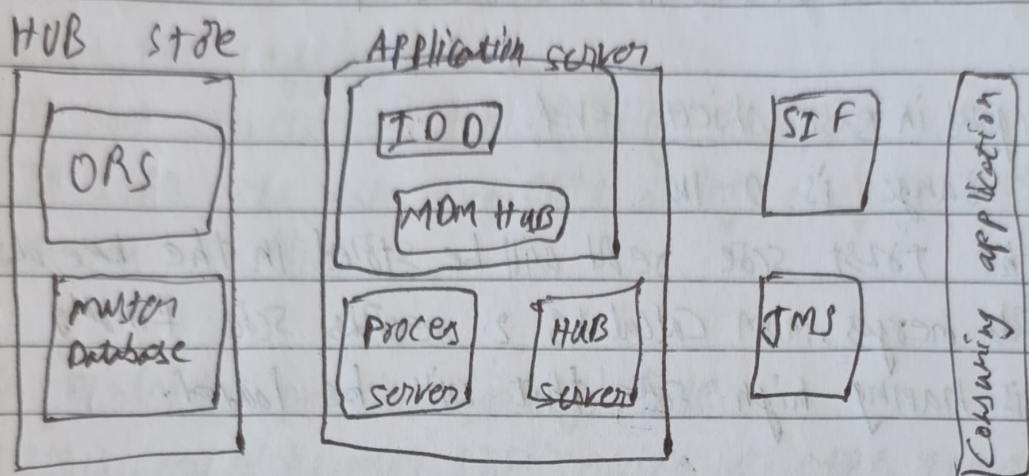
Trust:-

- * It is applicable for all columns in Base object for which trust is enabled.
- * It is configured in Base object level.
- * Trust score range is 0-100
- * Based on the trust score record will be stored in the base object.
- * When 2 records merges, MDN calculates 2 records score finally which one is having high score that will be loaded.

Validation Framework:-

- * It works on the validation rules configured for the base objects.
- * Rules are applicable for columns which are enabled for the option validate in the base object.
- * Types of validations:- Existence check, Domain check, Referential integrity, pattern validation & custom.
- * While configuring validation rule, you need to provide the downgrade percentage, if the validation rule met, & enable reserve minimum trust
- * Amount to downgrade by is a % of trust.
- * Reserve minimum trust, if enabled then trust score of the attribute will not downgrade below the minimum trust score configured when the validation rule is met.

Architecture :-



* Only one master Database also known as CMX-system

Batch Jobs :-

1. Landing Jobs :-

* Extract & Load

* ETL Tools - Info'matic ETL, Ab Initio

2. Stage Jobs :-

* Records from landing to stage

* Cleaning

* Standardization

* Delta Detection

* Reject Handling

* HDD

3. Load Jobs:-

- * Records from stage to Bo
- * Trust Calculation.
- * Reject Handling.

Tokenization:-

- * It's done automatically.
- * It's done end of Load Job (and / or) start of match Job.
- * It won't done tokenization for encrypted Data.
- * Generally we execute batch Job & Tokenization Job separately.

HUB State indicator (Hub-STATE-IND):-

- 1 → Active
- 0 → Pending
- 1 → Deleted
- 9 → Deleted present in XRef table.

Consolidation indicator:-

- 4 → New Load with merge
- 3 → Ready for match
- 2 → Ready for merge
- 1 → Consolidated (unique record (best version of truth))
- 9 → on hold.

Survivorship Rules:-

1. Trust score
2. Last update date (most recent one)
3. Rowid Object (Highest rowid value)
4. Rowid xref.

Hierarchy Basics:-

- * Entities
- * Entity Types
- * Hierarchy
- * Hierarchy Type
- * Relationship Objects
- * Relationship Type

Match:- It is a process of deduplication

* There are 2 types of matches

1. Exact:-

- * No need to tokenization
- * It's goes to MTCH table

2. Fuzzy:-

- * Tokenization $\rightarrow$ STRP table, DRTY table $\rightarrow$ match process $\rightarrow$ MTCH table with auto merge as D.